

Choosing the Right Statistical Test

One-sample t-test · Mann–Whitney U · Chi-square test of independence



t-test



Mann–Whitney U



Chi-square

Three tests, one underlying logic



One-sample t-test

Sample mean BMI vs. a fixed WHO reference value

Continuous data, approx. normal



Mann-Whitney U

Attachment loss (CAL) between smokers and non-smokers

Continuous data, NOT normal



Chi-square test

Periodontitis status vs. diabetes status

Two categorical variables

Null hypothesis significance testing (NHST)

The same 4-step logic runs underneath every test you'll use today.

1

State H_0

Usually “no difference” or “no association”

2

Pick a statistic

That matched to your data type (continuous/normal, continuous/skewed, or categorical respectively)

3

Compute the p-value

$P(\text{this extreme or more} \mid H_0 \text{ true})$

4

Compare to α

Conventionally 0.05 — decide to reject H_0 or not

Why report a confidence interval at all?

A p-value collapses everything into one yes/no threshold. A CI keeps the direction, size, and precision of the effect.



CI adds direction + magnitude

On top of the bare significance decision, the CI tells you which way the effect points and roughly how big it is.



Point estimate stays your best guess

Regardless of how wide the CI is, the point estimate remains the single best approximation of the true value.



“Values outside the CI are excluded”

is a common but incorrect way to read a confidence interval. A 95% CI isn't a hard cutoff between possible and impossible values, it's a range of values that are reasonably compatible with your data. Numbers just outside the interval aren't ruled out; they're just slightly less compatible with what you observed, and that compatibility fades gradually rather than snapping off at the boundary.



Significant p = clinically important

Wrong — statistical significance depends on sample size too; tiny, meaningless effects can be “significant” in big samples.

From: **Noninferiority of Silver Diamine Fluoride vs Sealants for Reducing Dental Caries Prevalence and Incidence: A Randomized Clinical Trial**

JAMA Pediatr. 2024;178(4):354-361. doi:10.1001/jamapediatrics.2023.6770

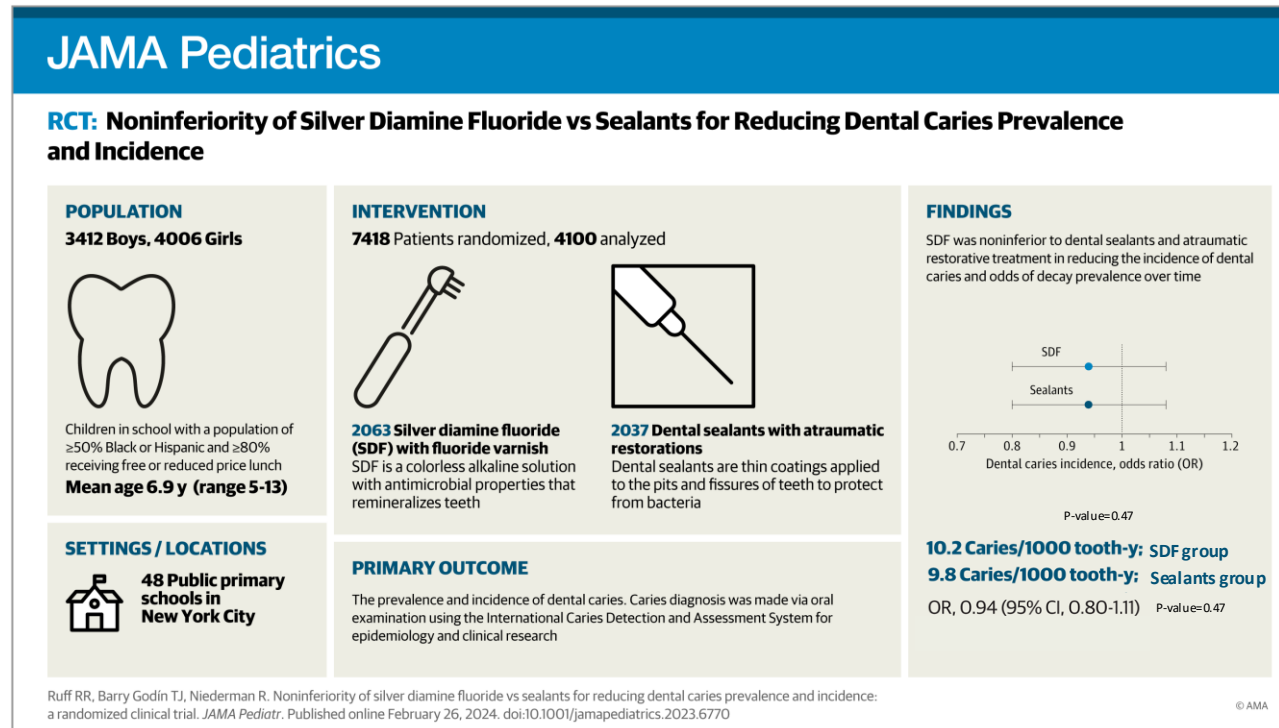


Figure Legend:

Silver Diamine Fluoride to Reduce Dental Caries Incidence

One-sample t-test — is average BMI off-target?

The question

Does our NHANES sample's mean BMI differ from the WHO reference value of 21.7?

There is only ONE sample here — you're comparing NHANES' mean against a fixed known constant, not against a second group.

Formula

$$t = \frac{(\bar{X} - \mu_0)}{\frac{s}{\sqrt{n}}}$$

t = t statistic

$\bar{X}$ = sample mean BMI

μ_0 = reference weight

S = standard deviation of the sample

n = sample size

$\frac{s}{\sqrt{n}}$ = standard error

Required: BMI must be approximately normally distributed (checked in Practical 3).

$t > t_{crit}$ we can reject H_0

Reading the SPSS output

Mean BMI	24.3
<u>t statistic</u>	8.46
df	119
Sig. (2-tailed)	< 0.001
95% CI	(23.7, 24.8)

Since the 95% CI (23.7–24.8) doesn't include 21.7, and $p < .001$, we reject H_0 : the sample mean BMI differs from the WHO reference.

Mann–Whitney U — comparing CAL by smoking status

Why not a t-test here?

In Task 1 we already showed `cal_mean` is NOT normally distributed. The t-test assumption is broken — so instead of comparing means directly, Mann–Whitney ranks ALL observations from both groups together and compares the ranks.

Non-smokers



Smokers



Skewed, non-normal shapes like these are exactly why we rank instead of average.



Careful with “it compares medians”.

Strictly, Mann–Whitney tests whether a random value from one group tends to be larger than one from the other (a difference in distributions). It's only exactly a median comparison when both groups' distributions have a similar shape.

EXERCISE 3

Chi-square test — periodontitis and diabetes, related?

Are these two categorical variables independent, or associated?

	Diabetes +	Diabetes –
Perio +	55	45
Perio –	30	70

$$\chi^2 = \sum (\text{Observed} - \text{Expected})^2 / \text{Expected}$$

df = (rows – 1)(cols – 1) = 1 for a 2×2 table

Before you trust the result...

Check the EXPECTED counts (not observed) in your SPSS crosstab.



All expected counts ≥ 5

→ Use Chi-square test



Any expected count < 5

→ Use Fisher's exact test

Rejecting H_0 only shows an association — not which direction causes which, and not whether a third factor explains both.

HOLD ON...

Wait — does that χ^2 result mean causation?

A classic (true!) example of why “associated” $\neq$ “caused by”:

“People who drink coffee have more lung cancer diagnoses”

This could be concluded in a sample when applying a statistical test of association



The tempting (wrong) reading

“Coffee drinkers $\rightarrow$ lung cancer $\rightarrow$ campaign to reduce coffee consumption

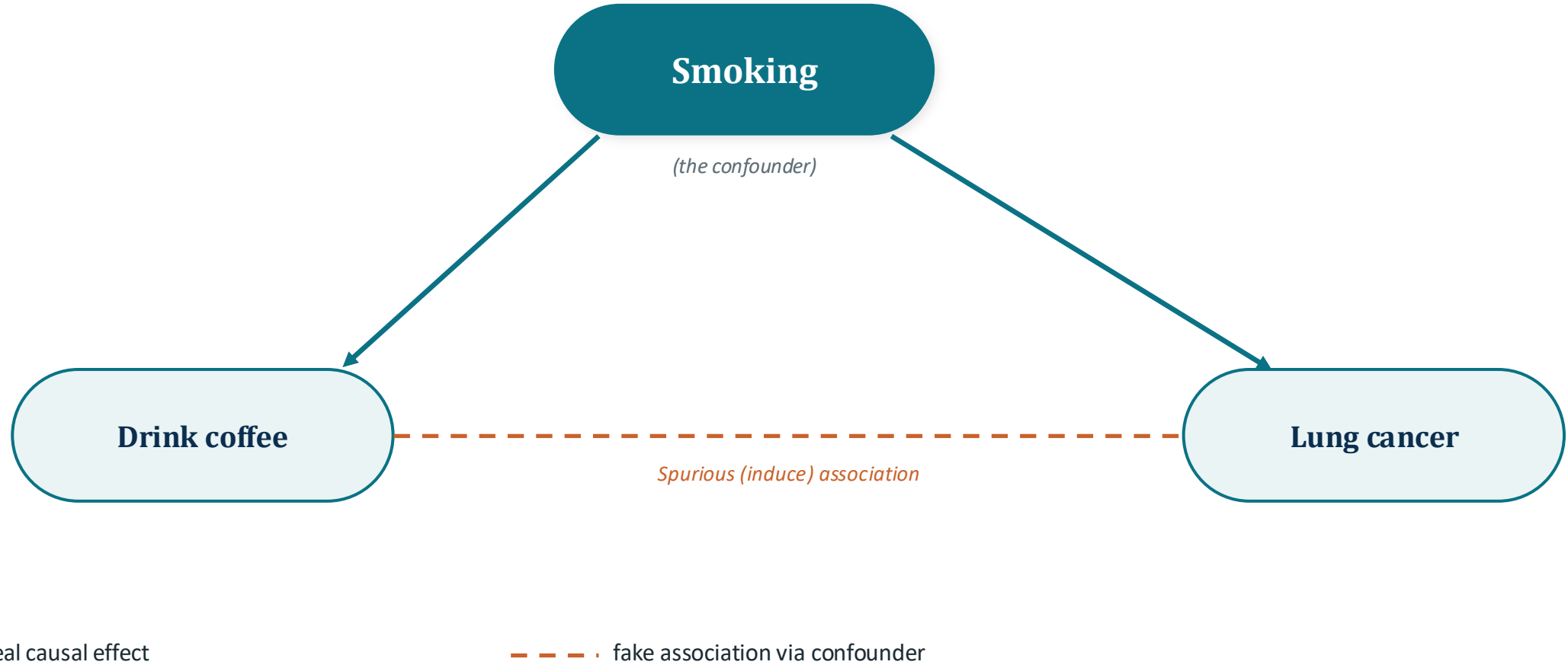


What's actually going on

Smoking explains both: coffee drinkers may smoke more AND have a higher risk of developing lung cancer.

This is a **CONFOUNDER**: a common cause of both the “exposure” (drink coffee) and the “outcome” (lung cancer) that creates a fake association between them.

The confounding triangle



The fix: adjust for (control for) smoking, and the coffee/lung cancer association disappears.

Three things to remember



p-values answer a narrow question

“How surprising is this data if H_0 is true?” — not “is my hypothesis true?”



Test selection follows data type + shape

Continuous & normal $\rightarrow$ t-test. Continuous & skewed $\rightarrow$ Mann–Whitney. Categorical $\rightarrow$ chi-square (or Fisher's).



Association $\neq$ causation

Always ask: could a confounder (like smoking) be driving both variables?

Questions? Bring them to the SPSS lab — let's run the numbers together.